

NATIONAL INSTITUTE OF TECHNOLOGY PATNA
Department of Civil Engineering
Mid Semester Examination Jan-June 2023

Course: B. Tech.
Semester: IV
Max. Time: 2Hrs

Subject: RCC Design
Course Code: CE43108
Max. Marks: 30

Note: Attempt all questions. Each question carries equal marks.
*IS Code is Not allowed. Assume grade of concrete **M30**, and Steel of **Fe415** grade.*

1. Illustrates the different design philosophy for RCC structures. 0.15⁴
2. Draw the stress-strain diagram of steel and concrete marking with permissible limits assumed in working stress method and limit state method.
3. Analyzed the beam of cross section given in Figure 1 by working stress method and find the moment of resistant of the beam.
4. Design a singly reinforced concrete beam to resisting a maximum bending moment of 50 kN-m using working stress method. 0.15⁴
5. Analyzed the beam of cross section given in Figure 1 by limit state method and find the moment of resistant of the beam.

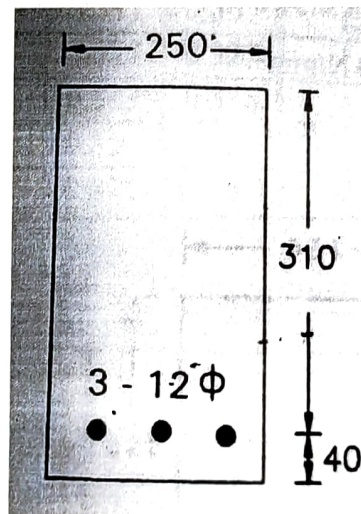


Figure 1.